

Solution Requirements

Date	30 September 2026
Team ID	6ab3ec17b5170a9b58b3a66b
Project Name	Fit-Buddy
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	User ID is collected for profile identification and persistence; no login credential flow is specified.	Medium
2	Authorization levels	Coach/admin view can access registered users and plans; role authentication is not specified.	Medium
3	External interfaces	Google Gemini API generates workout and nutrition/recovery guidance; SQLite stores application data.	High

		Third-party AI service for personalized workout plans and nutrition/recovery guidance.	High
4	Transactions processing	No payments are involved. Validated fitness inputs are sent to Gemini; generated plans and guidance are stored in SQLite.	High
5	Reporting	Result page displays the 7-day plan and nutrition/recovery tip; coach/admin view lists users and stored plans.	Medium
6	Business rules, etc.	Generate a 7-day plan from goal, age, weight and intensity; retain the original plan and store a revised plan after feedback.	High
7	Compliance to laws or regulations	API key is supplied through environment configuration. No specific regulatory standard is documented in the project files.	Medium
8	<i>Other</i>	Accept user feedback and display an updated workout plan while preserving the original plan.	High

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	Responsive local API behavior is expected under normal usage; no benchmark is specified.	Normal local use; no numeric benchmark specified.
2	Scalability	Modular FastAPI and SQLite design can be extended to hosted databases and larger deployments.	Extensible design; no user/concurrency target specified.
3	Security & Data Privacy	Keep the Gemini API key in environment configuration and validate user input before processing.	No hard-coded API key; validation and error handling present.
4	Reliability & Availability	Handle API/database errors and return usable application responses during local testing.	Successful local testing; no uptime target specified.
5	Usability & Accessibility	Provide a simple browser form and readable Jinja2 result pages for the workout output.	Core flow usable through a browser; no WCAG target specified.
6	Other	Use modular Python files for routing, schemas, database and AI generation.	Maintainable modular structure; portable local Python setup.